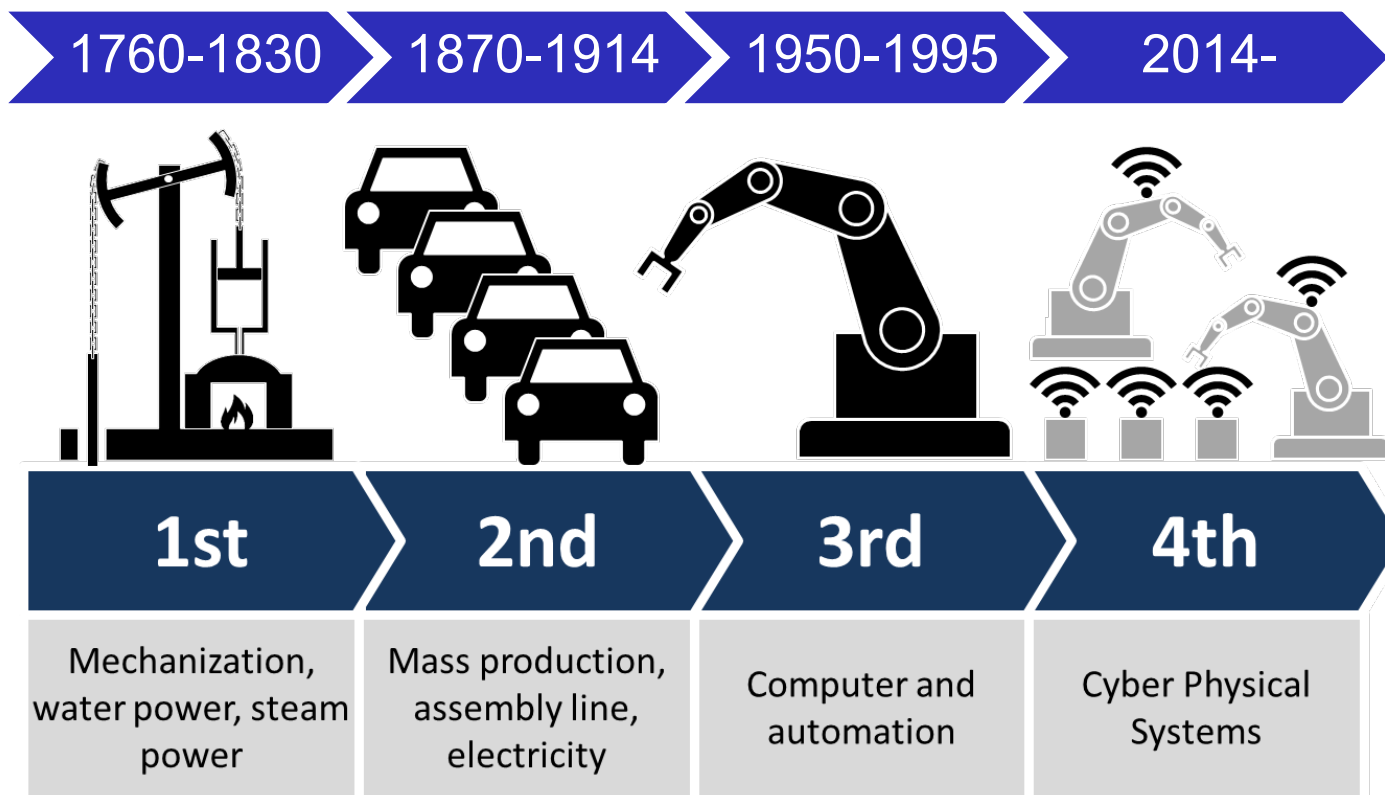




Smart Manufacturing

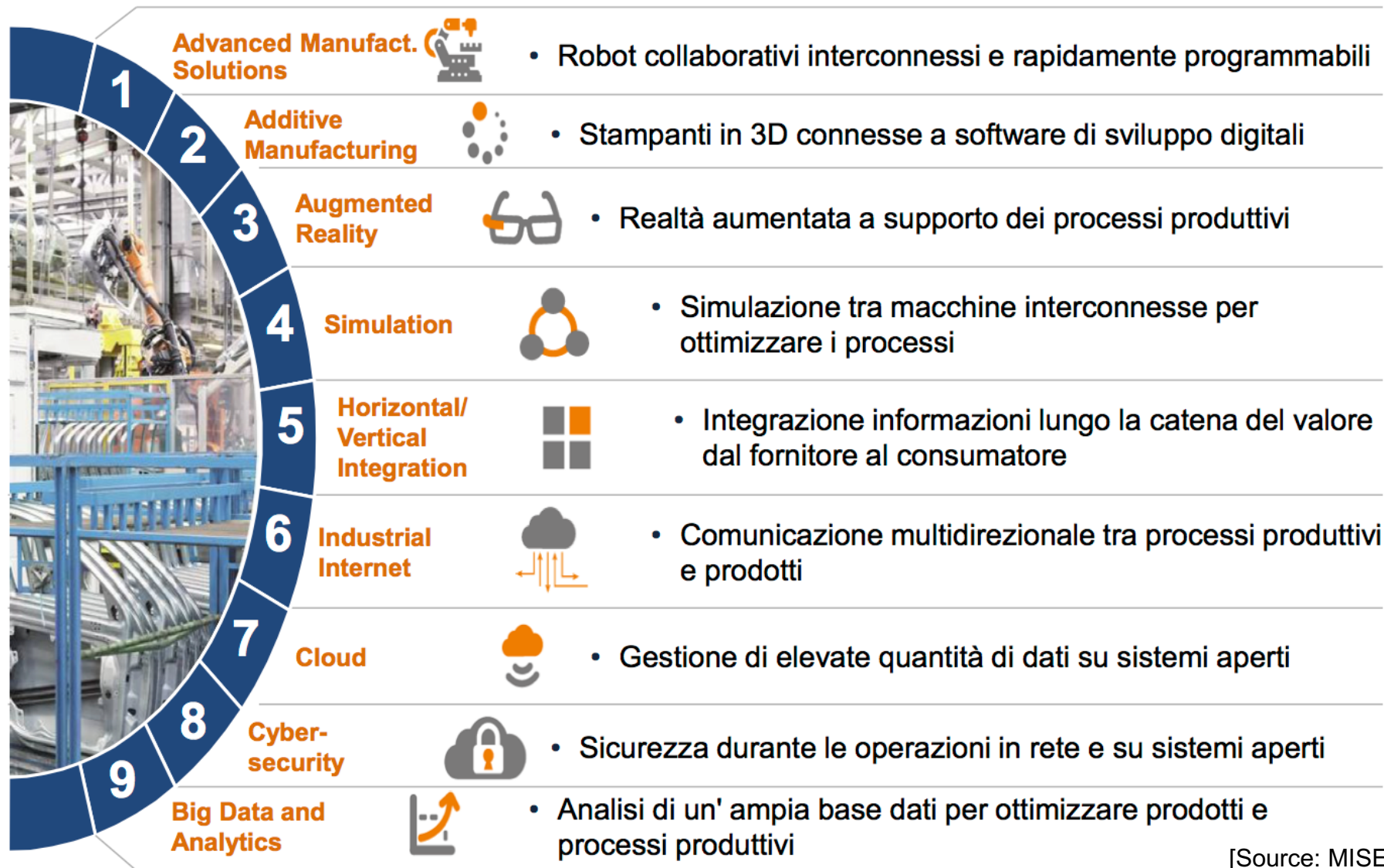


Other names used in different countries:

- Smart Manufacturing
- Smart Factories
- Industrial Internet of Things



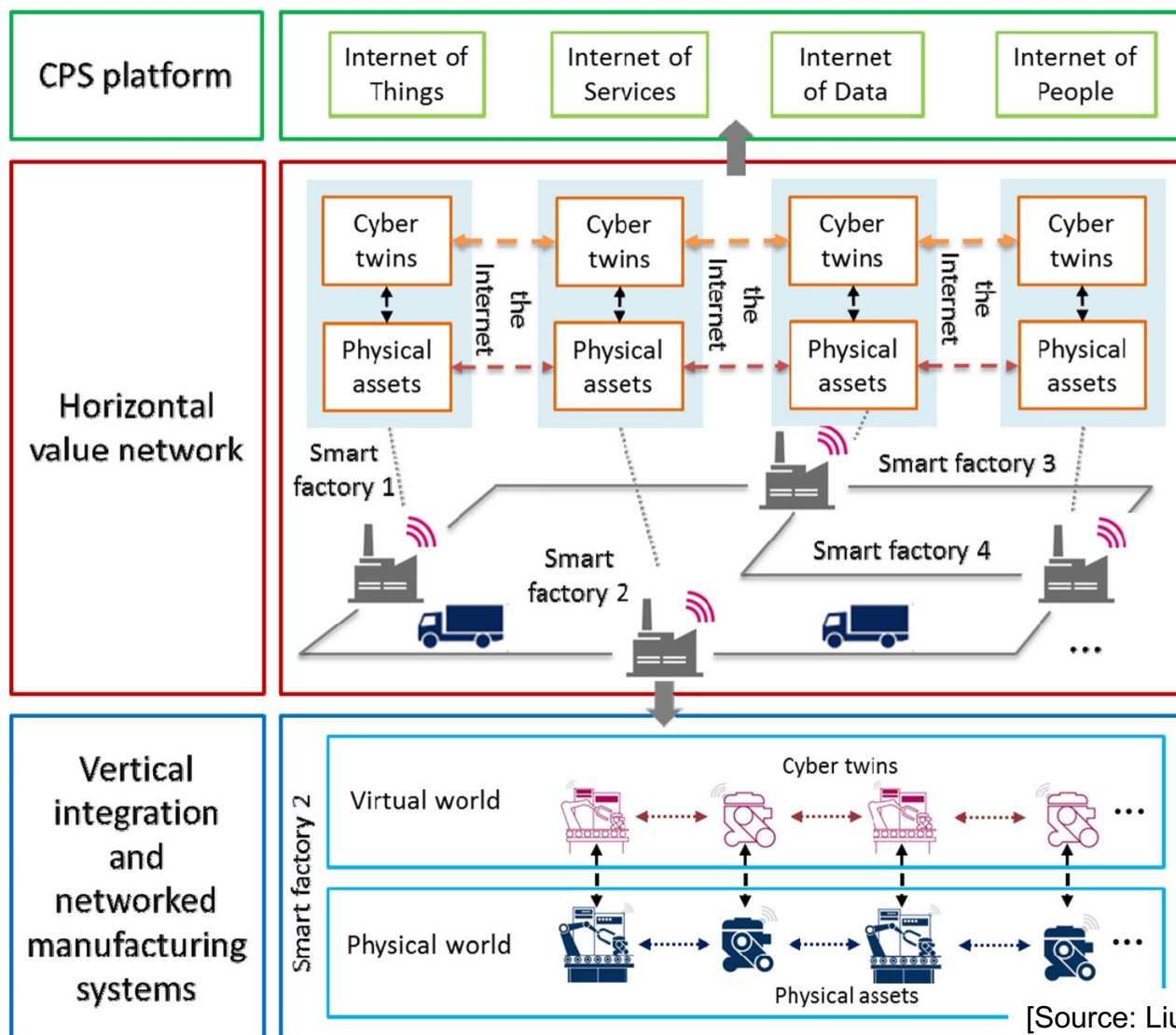
ITALIAN INDUSTRY 4.0 ENABLING TECHNOLOGIES

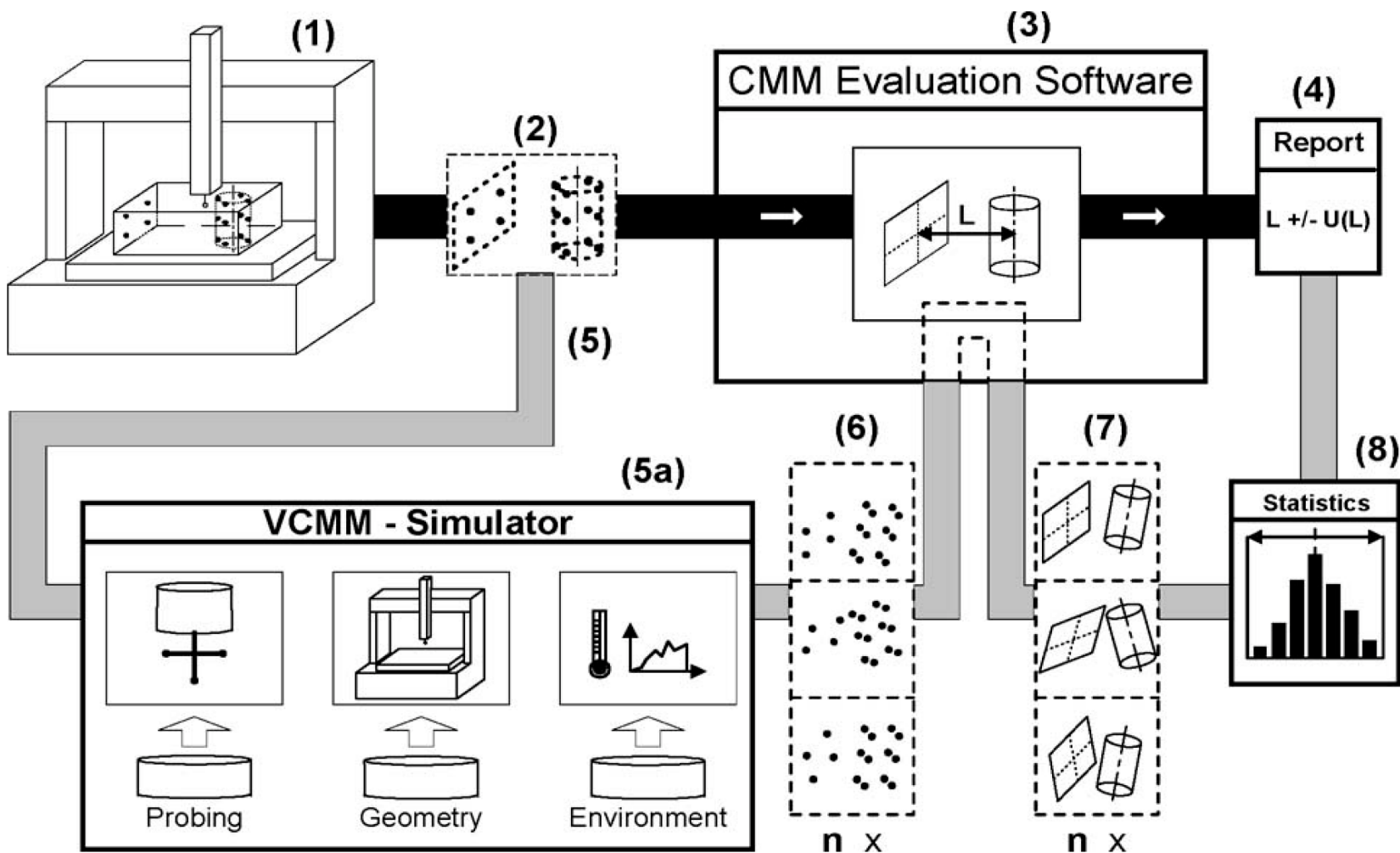


[Source: MISE]



HORIZONTAL AND VERTICAL INTEGRATION





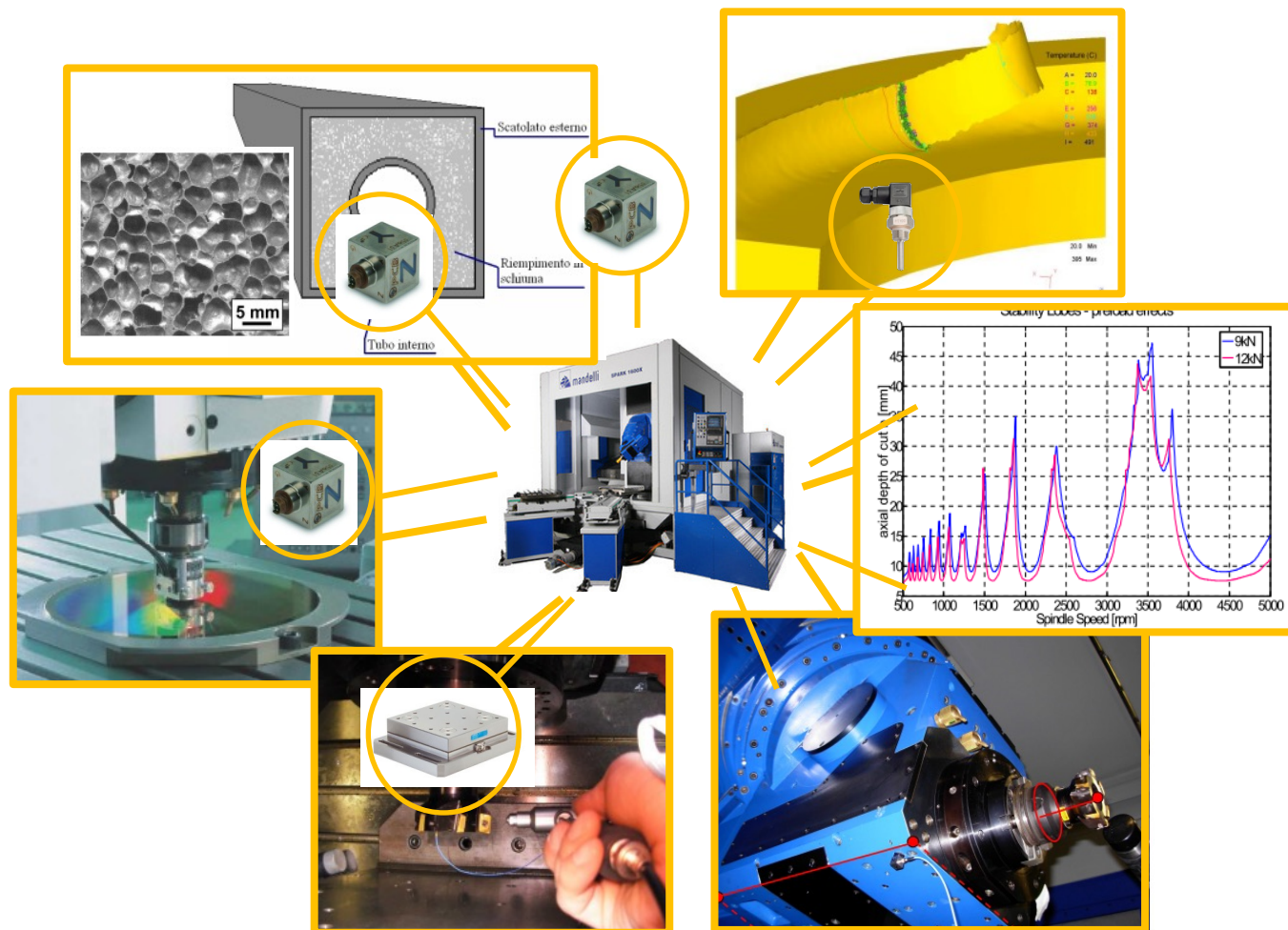
VCMM = Virtual Coordinate Measuring Machine



ADVANCED MANUFACTURING SOLUTIONS

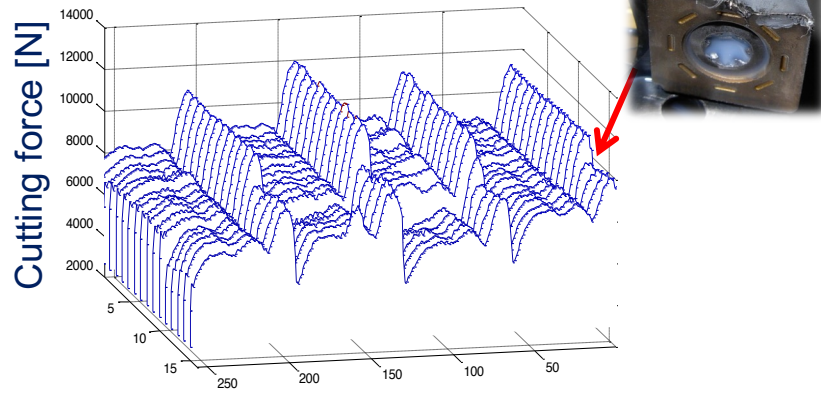
Machine Tools and Sensors:

- Power
- Force
- Vibrations
- Acoustic emission
- Images
- Videos (IR/visible range)

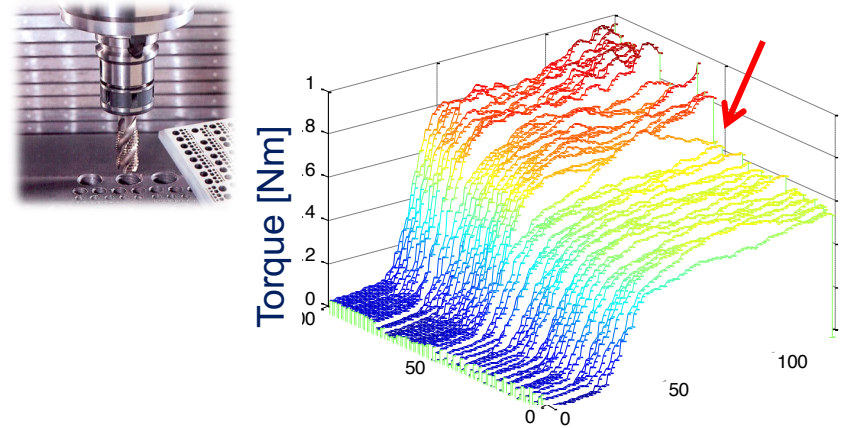


[Source: Polimi]

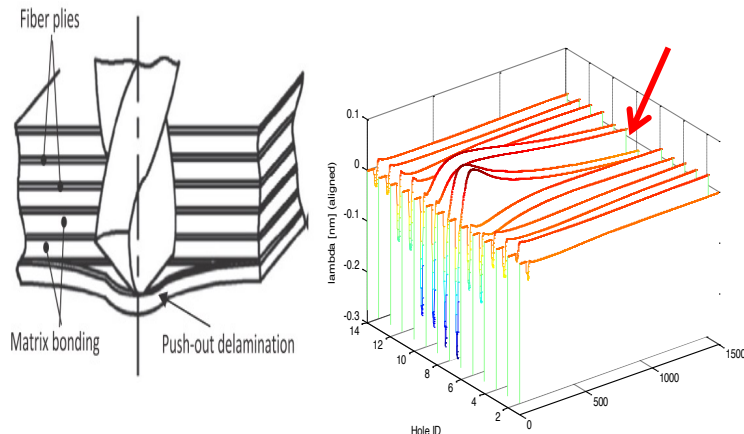
Milling hard-to-cut materials



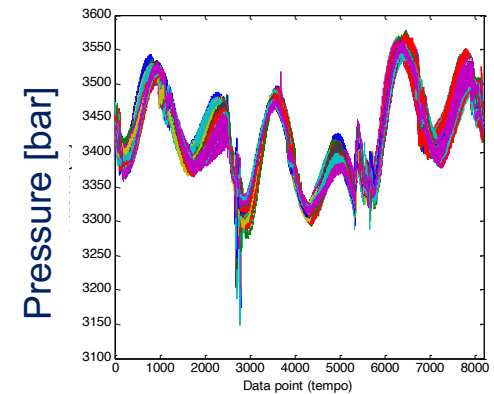
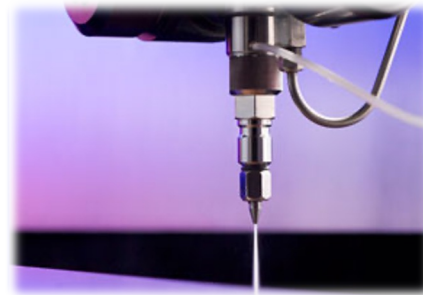
Tapping monitoring



Drilling of multi-layer materials (aerospace)



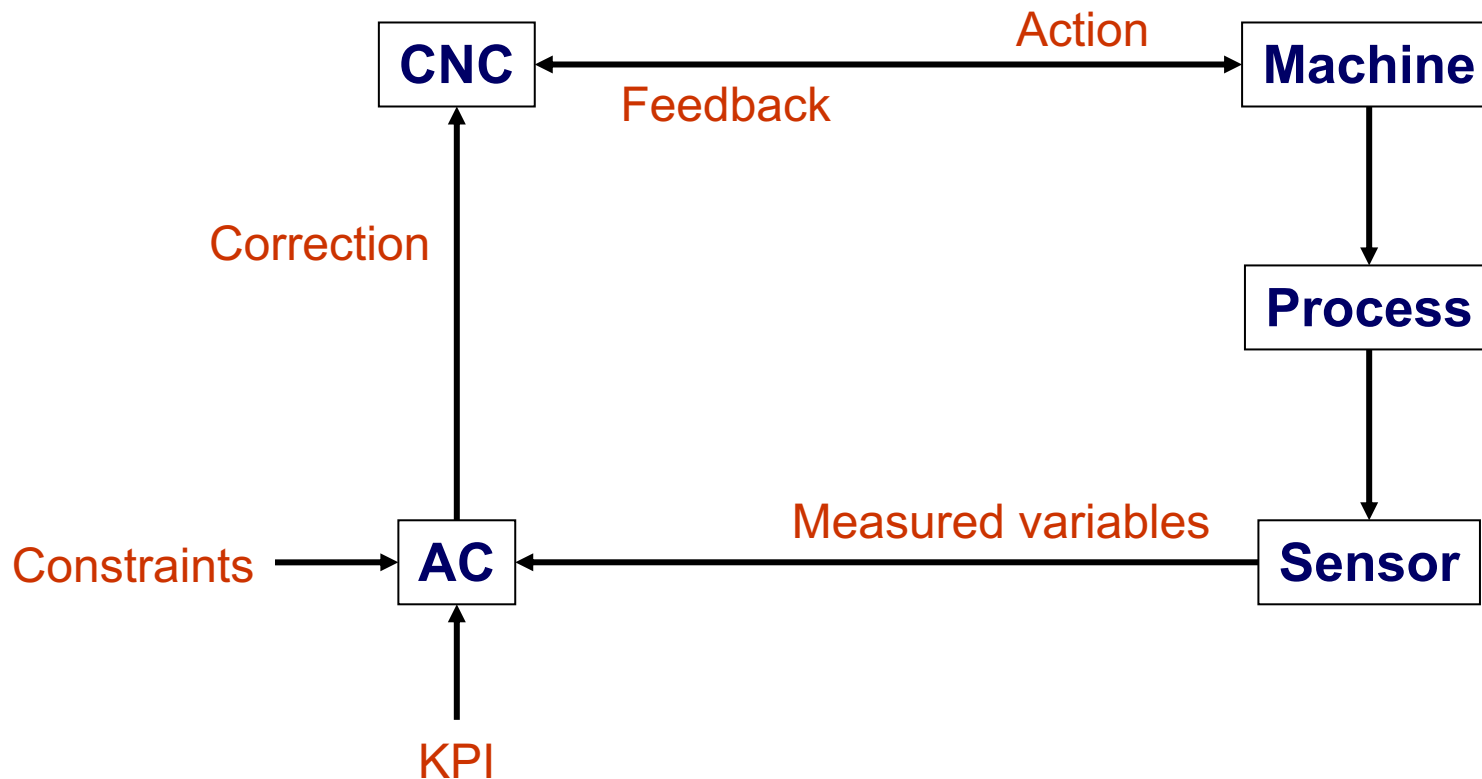
Waterjet and Abrasive Waterjet cutting



[Source: Polimi]



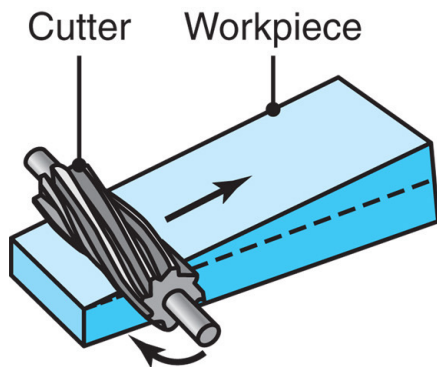
Control scheme that adjusts the control system parameters from conditions detected during the process (ISO 2806:1994).



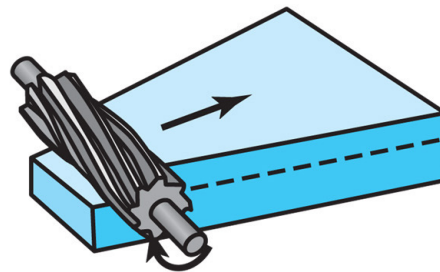


ADAPTIVE CONTROL - EXAMPLE

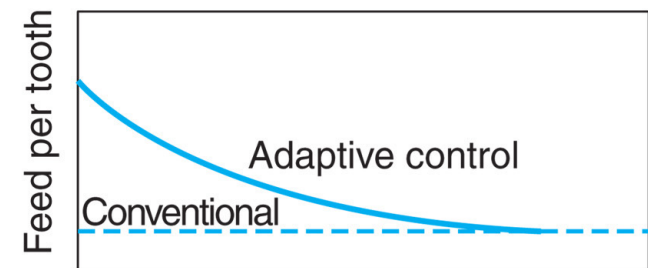
Adaptive control needs in slab milling.



(a) Variable depth of cut



(b) Variable width of cut

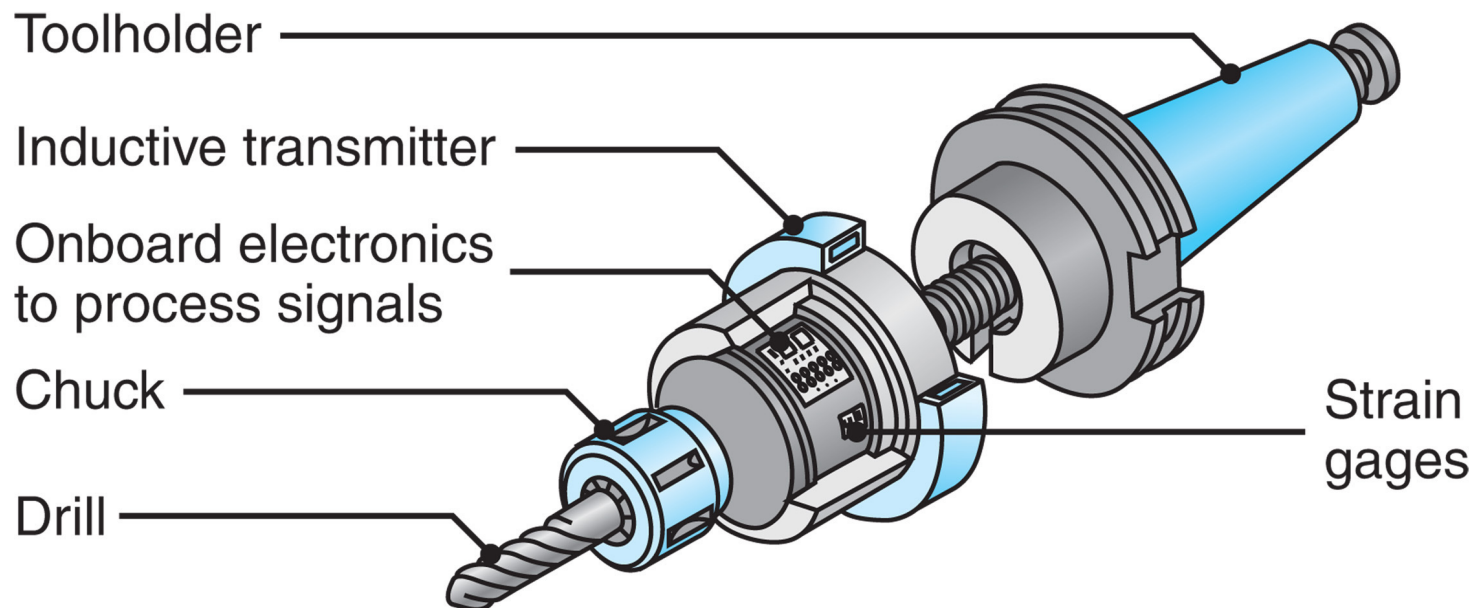


(c) Cutter travel

ADAPTIVE CONTROL AND SENSORS

Sensor

Unit which is actuated by a physical quantity and which gives a signal representing the value of that physical quantity (ISO 2806:1994).



A toolholder equipped with thrust-force and torque sensors (*smart toolholder*).



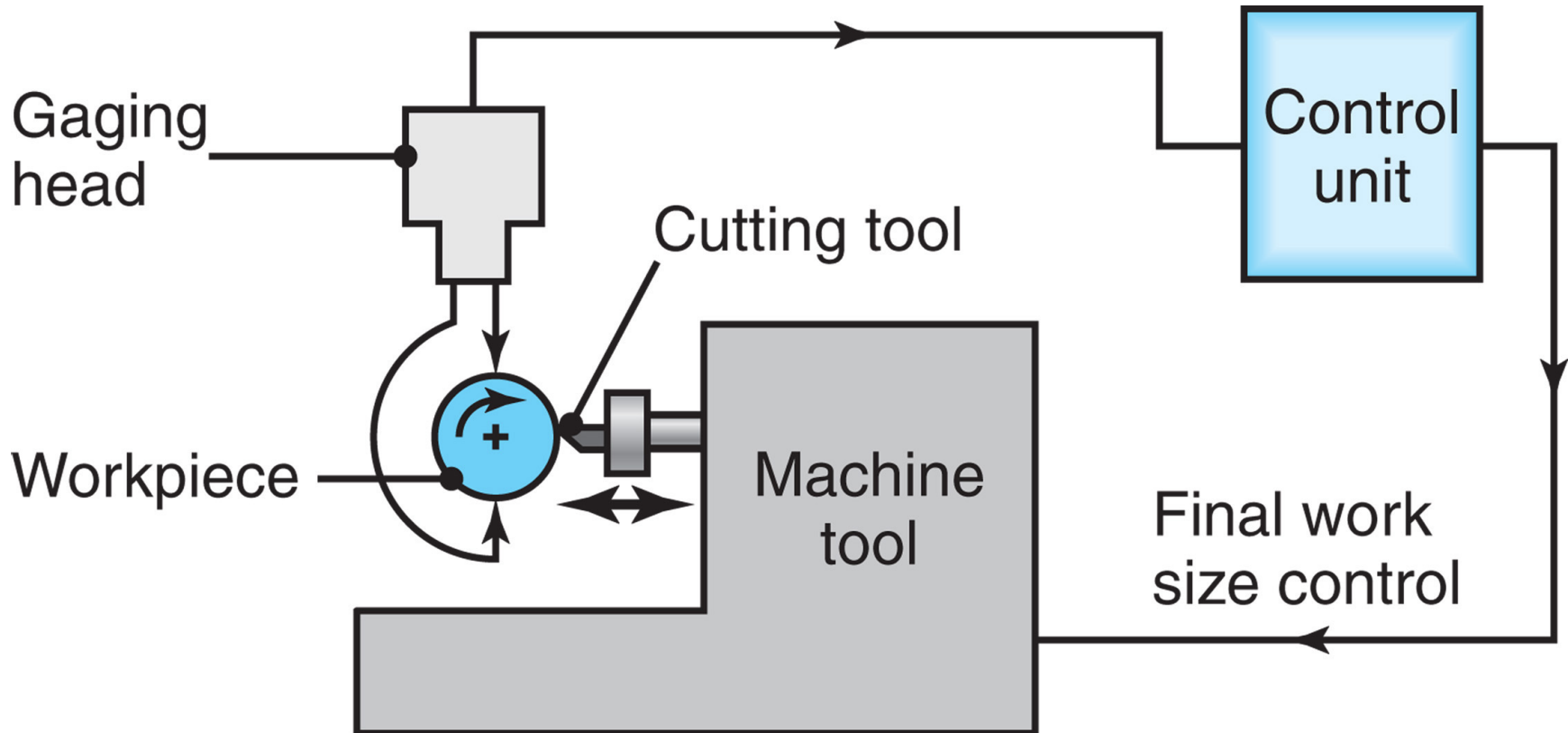
Control scheme that adjusts the control system parameters from conditions detected during the process (ISO 2806:1994).

- GAC - geometric adaptive control
- ACC - adaptive control constraint
- ACO - adaptive control optimization

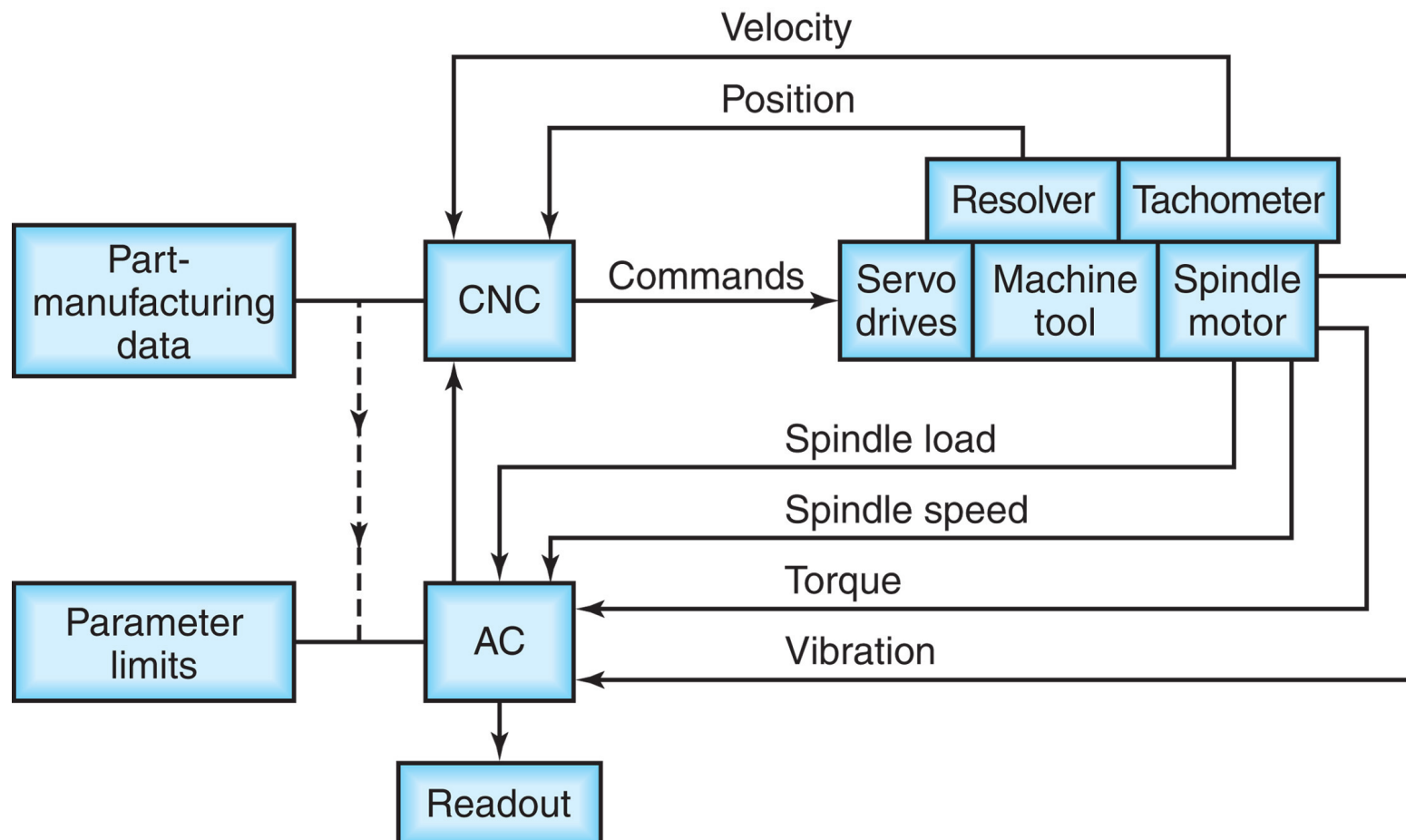


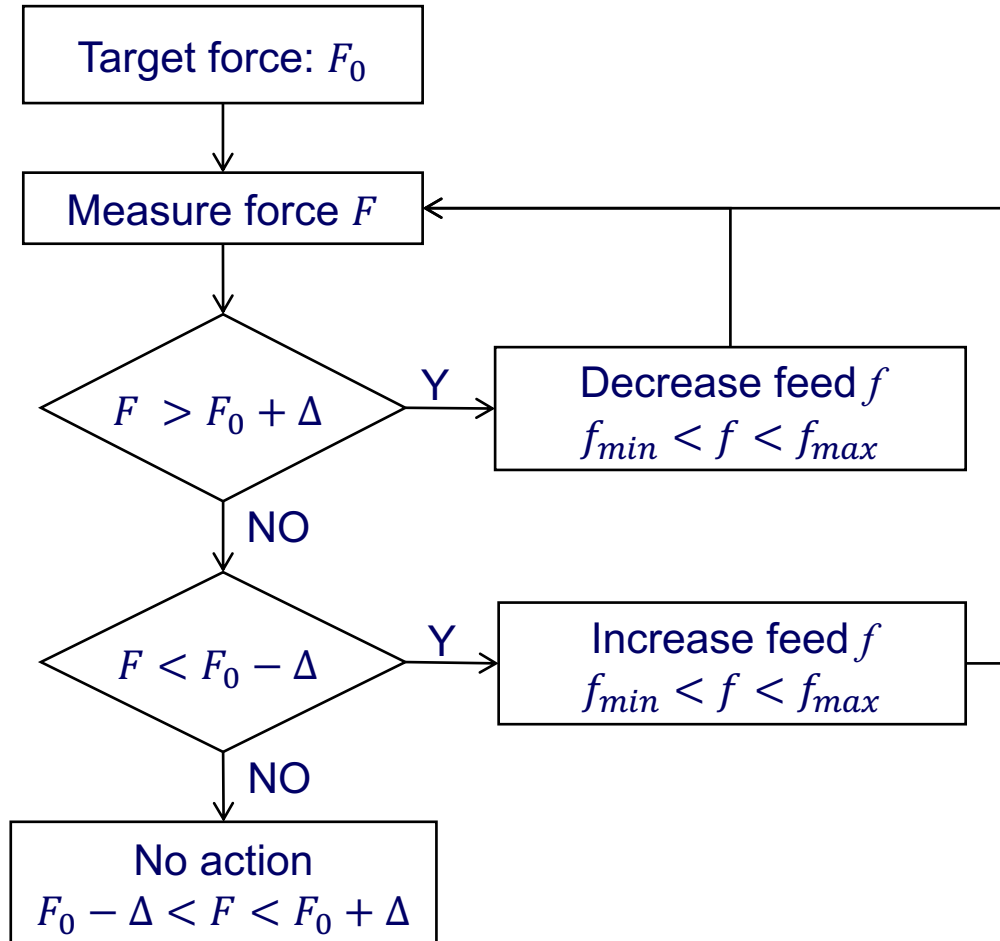
GEOMETRIC ADAPTIVE CONTROL

GAC aims at keeping tolerances and surface finishing in given ranges.



In-process inspection of workpiece diameter in a turning operation.





ACC aims at keeping a technological variable in a given range satisfying the constraints of the technological system.

Typical variables are:

- Temperature
- Cutting Force
- Cutting Power
- Width of vibrations



ACO aims at machining at optimal conditions.

